



Aerodynamic optimization of hypersonic blunted waveriders based on symbolic regression

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ABSTRACT

The previous waverider optimization was mainly focused on the original configuration with sharp leading edge. In actual application, the leading edge must be blunted for hypersonic flight, which can change the aerodynamic performance significantly. Thus, the optimum waverider with sharp leading edge doesn't mean that it's still optimum after bluntness. To solve the problem, this paper first investigates the influence mechanism of leading edge bluntness on the aerodynamic performance of the waveriders in detail. And the novel methodology of symbolic regression is employed to establish an analytical pressure increment model for the original waverider caused by the bluntness effects. Then an efficient aerodynamic model for the blunted waverider is constructed by combining traditional approximate methods and the pressure increment model, which is further incorporated into the Genetic Algorithm optimization framework. Results show that the resulting blunted optimized waveriders have distinctly different shapes and better aerodynamic performance compared to previous optimization that considers only sharp leading edge. And as the bluntness radius or the design lift increases, the improvement of lift-to-drag ratio (L/D) turns larger. Finally, when the center of pressure is constrained during the optimization, the blunted optimized waverider exhibits both better trim characteristic and higher L/D .

1. Introduction

High lift-to-drag ratio (L/D) is a key design objective for various kinds of hypersonic vehicles in that higher L/D means larger down and cross range. For example, a hypersonic vehicle with the L/D of 3.5 would have more than twice the down and cross range capability of the Space Shuttle. This greater range can be used to significantly increase the operational flexibility of future reusable space systems, including the ability to reach demanding orbits and to quickly return for reuse [1]. However, it is well-known that high L/D is difficult to achieve at hypersonic flight conditions due to the severe wave drag and friction drag. Indeed, as the Mach number increases across the supersonic and hypersonic regimes, Kuchemann have pointed out that a type of “ L/D barrier” exists for traditional hypersonic vehicles [2].

To break this barrier, the concept of waverider has come to renaissance since the 1980s, which was first proposed by Nonweiler [3].

Differing from the flow physics around traditional configurations, the shock wave is attached to the entire leading edge of the waverider, thus preventing the leakage of high-pressure gas from the lower surface onto the upper surface and achieving excellent aerodynamic efficiency. Then sorts of waverider design methods from different types of generating flowfields were developed to improve the volumetric efficiency, payload, and flexibility [4–14]. And efforts on incorporation of the waveriders to practical hypersonic vehicles were also made for different aerospace missions [15–17]. A detailed overview of researches on waverider design methods was given by Ding et al. [18] and Zhao et al. [19].

The original waverider is generated with sharp leading edge that supports attached shock wave and minimizes the drag. However, in practical applications, the leading edge must be blunted for heating, manufacturing, and handling concerns [20]. Many researches have indicated that the leading edge bluntness can result in significant degradation of the aerodynamic performance of waverider, especially

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Nomenclature

CL	=lift coefficient
CD	=drag coefficient
CD_{INV}	=inviscid wave drag coefficient
CD_{VIS}	=viscous friction drag coefficient
Cmz	=pitching moment coefficient
d	=plate thickness or cylinder diameter, m
M_∞	=freestream Mach number
H	=flight altitude, km
L	=length of the waverider, m
P	=the surface pressure, Pa
ΔP	=the surface pressure increment, Pa
P_∞	=the freestream static pressure, Pa
R	=bluntness radius of the leading edge, m
Re	=Reynolds number

T	=temperature, K
U	=velocity along the X axis, m/s
U_∞	=freestream velocity, m/s
x	=distance measured from the nose in the flow direction, m
X_{cp}	=center of pressure, corresponding to the total length L
X_{cg}	=center of gravity, corresponding to the total length L
α	=angle of attack
γ	=ratio of the specific heats, 1.4 for perfect gas
δ_x^*	=boundary layer displacement thickness
β	=shock wave angle
θ	=deflection angle of body surface relative to the freestream direction
ρ	=density, kg/m ³
Λ	=local sweep angle of the leading edge
$\bar{\chi}$	=viscous interaction parameter

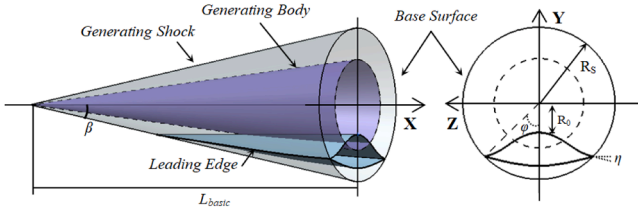


Fig. 1. Generation of cone-derived waverider.

the L/D [21–23]. Currently, the optimization of waverider is mainly focused on the idealized waverider with sharp leading edge, such as the inviscid optimization by Kim et al. [24], viscous optimization by Bowcutt and Corda et al. [5,25], and the strong-viscous-interaction optimization by Anderson et al. [26]. The progress of the above classical optimization demonstrates clearly that the more important physical effects are incorporated into the optimization framework, the more desirable waveriders with better performance can be obtained. Accordingly, in order to generate the optimum waveriders with blunted leading edge, it's necessary that the bluntness effects were taken into account during the optimization process.

Considering that the use of CFD simulations is extremely expensive and time-consuming, efficient aerodynamic models are generally preferred and widely used for the preliminary aerodynamic optimization of hypersonic vehicles, which have exhibited enough accuracy in several applications of waverider optimization with sharp leading edge [14,26]. For the blunted waveriders, a tough issue is how to accurately assess the bluntness effects, including both the aerodynamic forces of the blunted leading edge itself and the corresponding influence of the detached shock on the upper and lower surface. The former may be evaluated by use of the modified Newtonian theory, whereas the latter depends on many factors, such as the Mach number, flight altitude, angle of attack, leading edge radius, and the shape of the leading edge, etc. Several methods have been developed to deal with the influence of the detached shock. For instance, Lukasiwicz [27] derived the pressure approximation equations of blunt-nosed flat plates and cylinders utilizing the blast-wave theory. Dejarnette et al. [28] developed an empirical relation to calculate the pressure distribution over blunted cone based on the second-order shock-expansion method. Cheng et al. [29] made a theoretical study about flows under the combined influence of bluntness and viscous interaction. Then Boger et al. [30] extended this theory to treat yawed bodies. However, these existing theories are only suitable for simple configurations mentioned above and cannot be applied to calculate the pressure distribution of blunted waveriders with

streamlined lower surface. What's more, it is extremely difficult to directly derive analytical expressions based on traditional theory and model development approaches due to the complex geometry feature and flow structure.

The rapid development of artificial intelligence techniques (AI) today provides the opportunity for automatic model construction. The data-driven science using AI techniques such as machine learning to discover knowledge from data has rapidly expanded in recent years. One typical method is the symbolic regression (SR). Koza [31] first proposed SR based on Genetic Algorithm (GA) and obtained the accurate expression between the independent and dependent variables from a given data set. Since then, SR has been applied in several fields and achieved excellent results. Schmidt et al. [32] identified the analytical relations about the complex system dynamics. Weatheritt et al. [33] employed SR for turbulence model formulation and established effective non-linear constitutive stress-strain relationships, which could improve the accuracy of separated flow prediction. Xiong et al. [34] developed explicit mathematical expressions that could predict the variation of mechanical properties with important features by SR. Therefore, in light of the powerful ability for constructing physical models, SR is chosen here to formulate the influence of detached shock on the pressure distribution of waverider.

This paper is arranged as follows: Firstly, the influence of leading edge bluntness on the aerodynamic forces of different waveriders and its mechanism are analyzed in Section 2. After that, Section 3 employs SR to establish an effective explicit model, which can predict the influence of bluntness on the pressure increment of the original waverider. Subsequently, an aerodynamic model for the blunted waverider is constructed and incorporated into the optimization framework based on Genetic Algorithm (GA) in Section 4. Then Section 5 provides a detailed comparison and analysis of the resulting optimum configurations with different constraints. Finally, the conclusions are given in Section 6.

2. Influence of bluntness

2.1. Generation of the cone-derived waverider

The cone-derived waverider is selected as the research object of this paper. Given a basic conical flowfield, the waverider can be determined by the base curve on the base plane. The base curve is determined by the following third-order polynomials:

$$y = \begin{cases} a + bz^2 - cz^3 (z < 0) \\ a + bz^2 + cz^3 (z \geq 0) \end{cases} \quad (1)$$

In order to depict the base curve more intuitively, define the design parameter $kw = R_0/R_s$, where R_0 is the Y-intercept of the base curve and

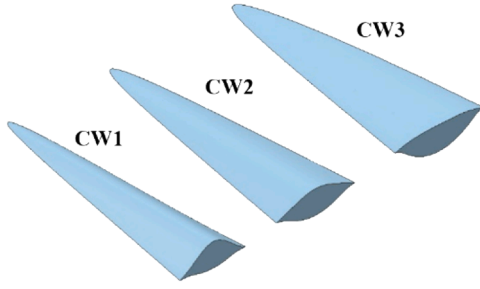


Fig. 2. Three typical waveriders.

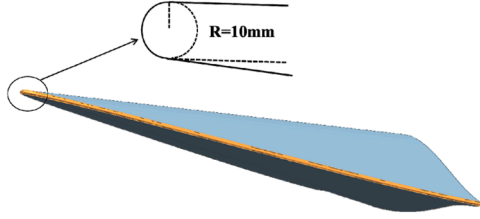


Fig. 3. Illustration of leading edge bluntness.

Table 1

Freestream condition for the DLR cylinder experiment[41].

P_∞ (Pa)	T_∞ (K)	u_∞ (m/s)	ρ_∞ (kg/m ³)	M_∞
687	694	4776	0.00326	8.78
ξ_{N_2}	ξ_{O_2}	ξ_{NO}	ξ_N	ξ_O
0.73555	0.134	0.0509	1.0e-9	0.07955

Table 2

Chemical reactions and reaction rate coefficients of the Gupta model[37].

o.	Reaction	A_{fr}	B_{fr}	C_{fr}	A_{br}	B_{br}	C_{br}
1	$O_2 + I_1 \rightleftharpoons 2O$ + I_1	3.61e18	59,400	-1	3.01e15	0	-0.5
2	$N_2 + I_2 \rightleftharpoons 2N$ + I_2	1.92e17	113,100	-0.5	1.09e16	0	-0.5
3	$N_2 + N \rightleftharpoons 2N + N$	4.15e22	113,100	-1.5	2.32e21	0	-1.5
4	$NO + I_3 \rightleftharpoons O + N + I_3$	3.97e20	75,600	-1.5	1.01e20	0	-1.5
5	$NO + O \rightleftharpoons O_2 + N$	3.18e9	19,700	1	9.63e11	3600	0.5
6	$N_2 + O \rightleftharpoons NO + N$	6.75e13	37,500	-1	1.50e13	0	0

 I_1 : O, N, O₂, N₂, NO; I_2 : O, O₂, N₂, NO; I_3 : O, N, O₂, N₂, NO.

R_s is the radius of the shock wave circle at the base surface [14]. In Fig. 1, the design parameter φ represents the azimuthal angles, and η represents the angles between the tangent line and Y axis at the point where the base curve intersects the shock wave. Then, given the three design parameters kw , φ , and η , the coefficients of Eq. (1) can be determined solely as follows:

$$\begin{cases} a = -kw \cdot R_s \\ b = \frac{\sin\varphi \tan\eta - 3\cos\varphi + 3kw}{R_s \sin^2\varphi} \\ c = -\frac{2\cos\varphi - \sin\varphi \tan\eta - 2kw}{R_s^2 \sin^3\varphi} \end{cases} \quad (2)$$

After the shock wave and the leading edge are determined, the lower surface can be generated by tracing the streamlines in the conical flowfield downstream from each point along the leading edge, and the upper surface is aligned with the freestream direction. All the cone-

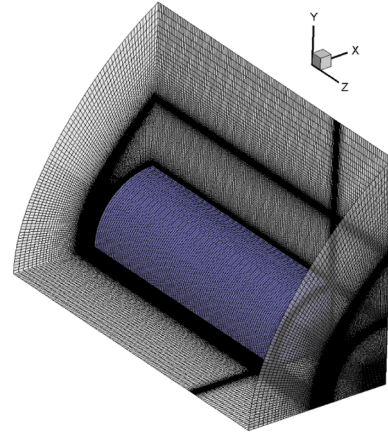


Fig. 4. Three-dimensional grid around the cylinder.

derived waveriders in the paper are finally scaled up to the same length $L = 4$ m.

In this paper, the influence of bluntness on the aerodynamic performance of the waverider is studied by taking the three typical waveriders as shown in Fig. 2. For the design condition $M_\infty=15$ and $\beta=8.5^\circ$, the design parameters kw and η are kept unchanged ($kw=0.4$, $\eta=20^\circ$), while φ is equal to 40° , 50° , and 60° for CW1, CW2, and CW3, respectively. It can be seen that these waveriders have different widths and leading edge distribution characteristics. In addition, the adding material method with circular tip radius suggested by Tincher [35] is adopted to blunt the leading edge, which is illustrated in Fig. 3.

2.2. Numerical methods

The steady-state laminar flow numerical simulations are performed in this paper. A cell-centered finite volume method is employed to solve the three-dimensional compressible N-S equations. The AUSM+ spatial discretization scheme is adopted, with an implicit lower-upper symmetric Gauss-Seidel scheme for temporal integration to accelerate convergence. Both the calorically perfect gas model and non-equilibrium reacting gas model are adopted and compared to examine the influence of the real gas effects on aerodynamic performance. For the chemical nonequilibrium simulations, a modified Fick's model is applied to calculate the species mass diffusion fluxes, ensuring that the sum of individual species' diffusion fluxes is zero [36]. The transport properties of mixture species are calculated according to Gupta's mixing rule [37], and the viscosity and thermal conductivity of species are calculated using collision cross-section data compiled by Wright [38]. More details about the CFD solver and its validation for hypersonic laminar with the calorically perfect gas model can be found in Refs. [39, 40].

For the nonequilibrium reacting gas model, the hypersonic cylinder flow experimental by Hannemann et al. [41] in DLR is selected as the validation case. The radius of the cylinder is 0.045 m. The freestream condition in the experiment, along with the gas species, is listed in Table 1. The freestream Reynolds number based on the radius is 2.11×10^{-4} and thus the fully laminar model is adopted. In addition, the chemical reaction model in numerical simulations is the 5-species gas and 6-reaction model developed by Gupta et al. [37], as given in Table 2. The above calculate methods are the same as those used in Ref. [41]. The three-dimensional computation is performed and the grid is shown in Fig. 4, consisting of $83 \times 152 \times 165$ grid points in streamwise, normal, and spanwise direction. Note that only one quarter of the windward side for cylinder is simulated due to the symmetry of the windward flow.

Fig. 5(a) shows the pressure contour at the symmetry plane around the cylinder, where the upper half is the chemical nonequilibrium result and the lower half is the perfect gas result. It is shown that the shock-off

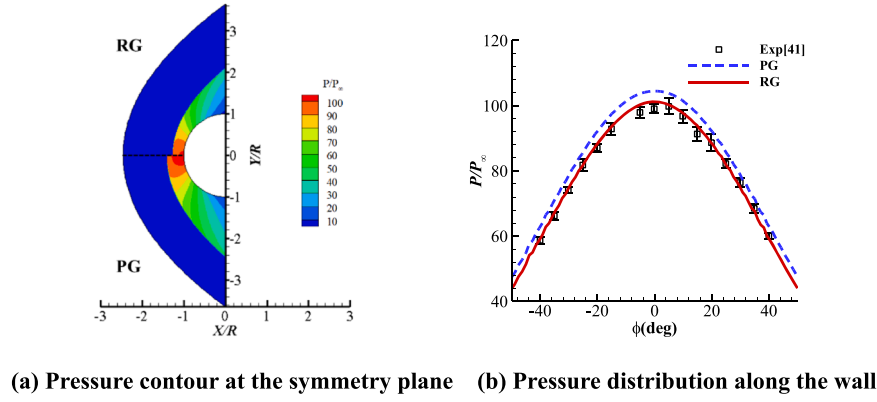


Fig. 5. Comparison of pressure from different methods.

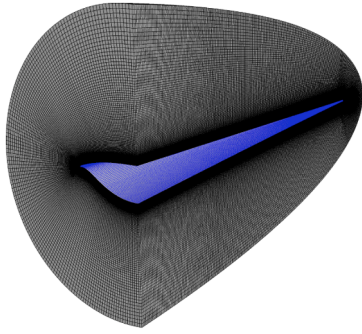


Fig. 6. Half-model grid of blunted waverider (6109,427 cells).

distance is remarkably reduced by the real gas effects. In addition, Fig. 5 (b) compares the pressure distribution along the wall, together with the experiment data [41]. It is found that the pressure distribution from the chemical nonequilibrium model agrees very well with the measured data, with the deviation of the stagnation pressure being only 2.22 %. And the deviation of the shock stand-off distance from the experiment (0.265R) is only 0.75 % for the current result (0.263R). The above results demonstrate that the current solver is reliable for hypersonic flow simulations with chemical reactions.

2.3. Grid independence validation

Two different half-model grids are generated to conduct the grid independence study. The number of the coarse and fine grid is about 6110,000 (Fig. 6) and 13,640,000 cells, respectively. The first grid spacing normal to the wall is 1×10^{-5} m, being small enough to satisfy $y^+ \leq 1$. The calculate condition is: $M_\infty=15$, $H=50$ km, and $\alpha=0^\circ$. The corresponding freestream Reynolds number is 1.2×10^6 , and thus the fully laminar model with an isothermal wall condition $T_w=1000$ K is adopted. For the chemical nonequilibrium simulations, the 5-species air and 6-reaction model (Table 2) are employed, along with the fully catalytic wall.

The corresponding force coefficients are listed in Table 3, including the laminar simulations with different gas models. Note that reference area of all the waveriders in this paper is 1.0 m^2 . It can be found that the

results from two grids are very close. In particular, the relative differences of CL , CD_{INV} , CD_{VIS} , X_{cp} between the coarse and fine grid are all less than 1 %. Therefore, the coarse grid is used for CFD simulations to save the computational costs.

2.4. Influence on aerodynamic performance

The waverider CW3 is taken here to study the influence of bluntness on aerodynamic performance, with the leading edge radius being 10 mm and 20 mm. The aerodynamic coefficients of CW3 blunted with different radii are plotted in Fig. 7, where *ORIGIN* stands for the original waverider with sharp leading edge. Not surprisingly, the aerodynamic performance of the waverider is significantly changed by the bluntness effects. Specifically, the lift decreases and the drag increases, accordingly reducing the L/D greatly. In addition, the X_{cp} also moves backward, which has vital influence on the trim characteristic.

In addition, Fig. 8 compares the relative variation of aerodynamic coefficient caused by bluntness, where CL and ΔCL in Fig. 8(a) denotes the lift coefficient of the original sharp waverider and the corresponding variation of the lift coefficient after bluntness, respectively. The maximum relative variation of aerodynamic coefficient after bluntness radius 10 mm is -9.25% , 12.97% , -19.67% , and 1.8% L , respectively. And the effect of the bluntness on aerodynamic performance decreases gradually as the angle of attack increases. However, with the increase of bluntness radius, the influence of bluntness increases significantly. When the bluntness radius is 20 mm, the maximum loss of L/D is about 35 %.

Furthermore, the comparison of the L/D and X_{cp} between the laminar perfect gas model and chemical nonequilibrium model is plotted in Fig. 9. It can be found that the chemical nonequilibrium effects have a slight influence on the L/D and X_{cp} at the flight condition, with the largest relative difference being only 1.82 % and 0.29 % for the L/D and X_{cp} (R20mm-RG, $\alpha=0^\circ$), respectively. And there is virtually no significant effect of chemically reacting flow on maximum L/D of waveriders for the current flight condition. Therefore, the laminar flow with the perfect gas model is employed for the following analysis of the paper.

The influence of the bluntness can be mainly decomposed into two parts: 1) the blunted leading edge; 2) the influence of the detached shock wave caused by the blunted leading edge on the upper and lower surface.

Table 3

Comparison of force coefficients at the condition: $M_\infty=15$, $H=50$ km, $\alpha=0^\circ$

	$CL \times 10^{-1}$		CD_{INV} ($\times 10^{-2}$)		CD_{VIS} ($\times 10^{-2}$)		X_{cp}	
	Coarse	Fine	Coarse	Fine	Coarse	Fine	Coarse	Fine
PG	1.156	1.154	2.094	2.090	1.663	1.663	0.646	0.646
RG	1.157	1.155	2.090	2.087	1.627	1.626	0.647	0.648

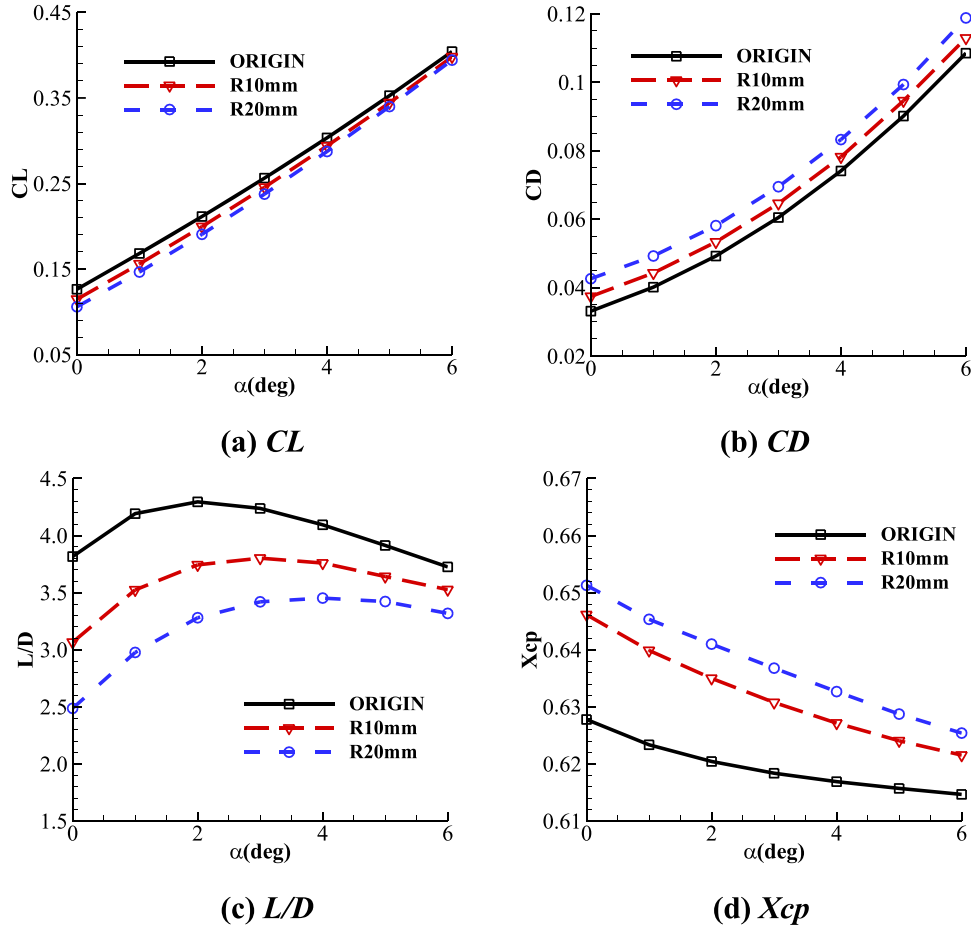


Fig. 7. Comparison of aerodynamic performance for CW3 with different bluntness radii.

The influence on the lift coefficient is presented in Fig. 10 for each part, where *Leading-edge*, *Upper+Lower*, and *Total_R10mm* corresponds to the blunted leading edge itself, the upper and lower surface without leading edge, and the whole blunted waverider, respectively. It's seen from Fig. 10 that the lift variation of the whole waverider (max: -9.25 %) is mainly attributed to the lift reduction of the upper and lower surface (max: -10.50 %), which is affected by the detached shock wave.

Furthermore, the influence on the drag coefficient is plotted in Fig. 11 for each part, including the inviscid wave drag and viscous friction drag, where CD , ΔCD_{INV} , ΔCD_{VIS} denotes the drag coefficient of the original sharp waverider, the corresponding variation of inviscid wave drag coefficient and viscous friction drag coefficient after bluntness. It's found that the wave drag increases dramatically (max: 12.88 %), which is mainly caused by the wave drag of the blunted leading edge (max: 16 %). And the detached shock leads to slight decrease in the wave drag of the upper and lower surfaces (max: -3 %). However, the total friction drag is almost kept unchanged after bluntness (max: 0.25 %), in that the friction drag of the blunted leading edge is almost counteracted by the friction drag reduction of the upper and lower surface.

In addition, the influence on the center of pressure is shown in Fig. 12, where *R10mm_NoLE* denotes the blunted waverider without including the blunted leading edge. Apparently, the variation of X_{cp} is mainly influenced by the forces on the upper and lower surface.

Based on the above analysis, for the blunted waverider, the variation of total lift and center of pressure is mainly dominated by the upper and lower surface, while the variation of total drag is mainly due to the wave drag of blunt leading edge itself.

2.5. Influence mechanism

The influence of the blunted leading edge itself on the aerodynamic characteristics is easy to understand. Thus, the following analysis is focused on the influence mechanism of the detached shock wave.

First, the pressure contour of the lower surface is given in Fig. 13 for the original and blunted waveriders. It's seen that after bluntness, the pressure first increases near the nose and then decreases. However, the region of the latter influence is evidently larger, leading to the reduction of CL and backward movement of X_{cp} .

The phenomenon of pressure increase near the nose can be easily explained along with the blast-wave theory [42]. And the more complex phenomenon of pressure decrease is called the pressure overexpansion effect, resulting from the interaction of the detached shock wave and the expansion waves. As illustrated in Fig. 14, some of the expansion waves intersect the sonic line and some intersect the detached shock wave. Those intersecting the sonic line are reflected as compressions, while those intersecting the shock wave are reflected as expansions. Then the surface pressure rises through the reflected compressions, and decreases through the reflected expansions [43]. The pressure distribution along the centerline at different angles of attack is further plotted in Fig. 15. It can be found that as the angle of attack increases, the pressure decrease near the nose is more evident and the position where the minimum pressure appears moves forward, indicating that the pressure overexpansion effect is enhanced.

Furthermore, the non-dimensional velocity contour near the body at different cross sections is displayed in Fig. 16 for the original and blunted waveriders. Interestingly, the detached shock wave makes the boundary layer thicker. This is because strong entropy gradients are generated in the nose region due to the detached shock. The entropy

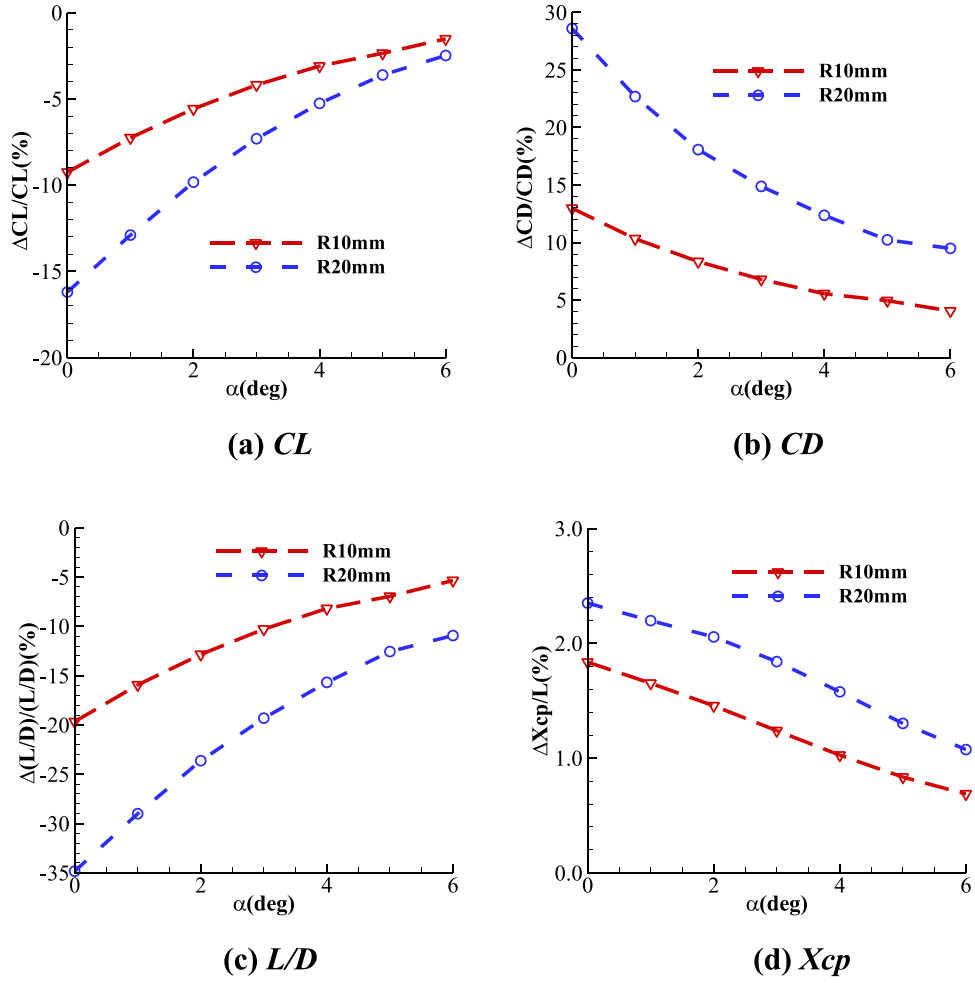
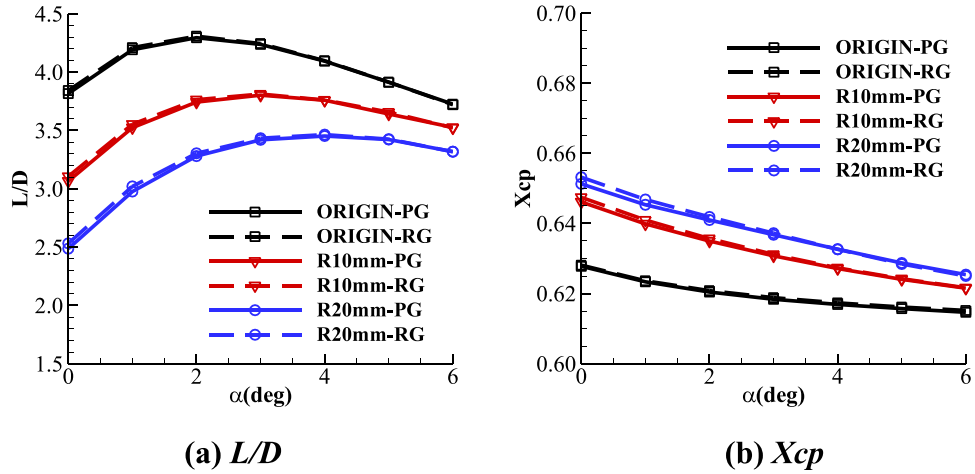


Fig. 8. Comparison of relative variation of aerodynamic coefficient after bluntness.

Fig. 9. Comparison of aerodynamic coefficient with different gas model, $M_\infty=15$, $H = 50$ km.

layer flows downstream and essentially effects that the boundary layer along surface grows for large distance from the nose [42]. As a result, the slope of the velocity profile inner the boundary layer, especially adjacent to the body, of the blunted waverider turns obviously lower than that of the original waverider, as illustrated in Fig. 17. Therefore, the shear stress and the corresponding friction drag of the upper and lower surface decreases significantly as shown in Fig. 11(b).

In fact, for the rapid evaluation of aerodynamic forces, the variation of the velocity profile is difficult to capture by traditional approximate models. However, based on the result that the total friction drag variation is very minor for the blunted waverider, we can assume that such variation can be neglected for the rapid calculation during the optimization. In order to verify this assumption, the drag variation for different flight conditions, bluntness radii, and waveriders is shown in Figs. 18-

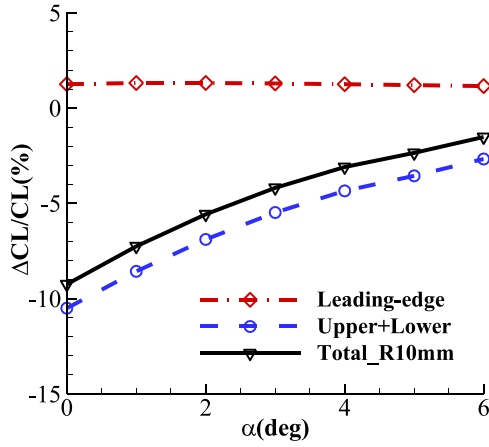


Fig. 10. Variation of lift coefficient after bluntness, $R = 10$ mm.

20. Note that *Total*, *INV*, and *VIS* denotes the total drag, the inviscid wave drag, and the viscous friction drag, respectively. The variation of the friction drag is less than 2.0 % for all the conditions, further demonstrating the reasonability of the assumption.

In addition, the pressure variation of the upper and lower surface by the bluntness is also difficult to evaluate by the existing aerodynamic models and cannot be neglected directly. The main objective of the following section is to solve this problem.

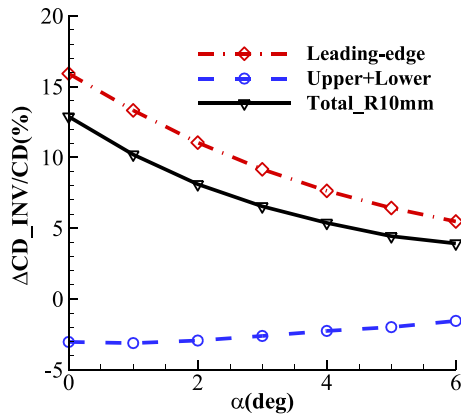
3. Methodology of symbolic regression

In cases where the function form is unknown, SR can find the best fitting function along with the relevant parameters and formulate mathematical models of complex systems using observed data. Therefore, SR is applied here to model the pressure increment of the upper and lower surface caused by the detached shock wave.

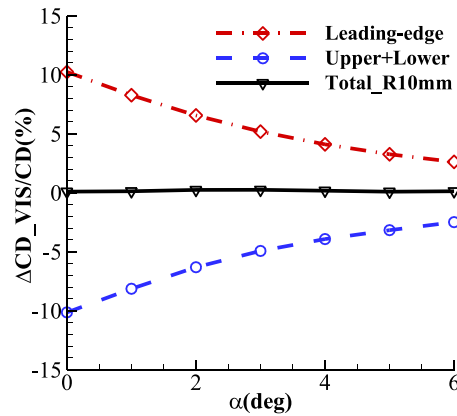
3.1. Geometry simplification

It is difficult to directly model for the waverider because the streamlines of the lower-surface vary with different configurations. In order to construct a universal aerodynamic model, the waverider is simplified as follows:

- 1) Generally speaking, the streamline of the cone-derived waverider is very close to a straight line. This can be observed in Fig. 21, where the streamline is distilled from the symmetry plane of CW3.



(a) Inviscid wave drag



(b) Viscous friction drag

Fig. 11. Variation of drag coefficient after bluntness, $R = 10$ mm.

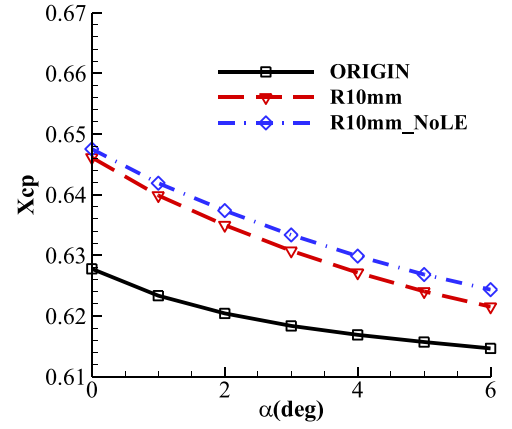


Fig. 12. Comparison of the center of pressure after bluntness.

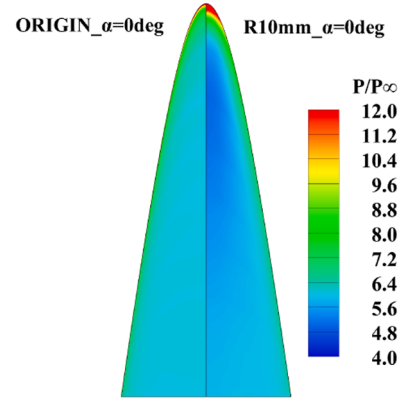


Fig. 13. Comparison of pressure contour.

Therefore, the streamline can be simplified as a straight line, meaning that the influence of the local slight curvature on the pressure increment induced by the detached shock wave is neglected.

- 2) Pressure is basically an impact event and is not affected much by the neighboring points [44]. Therefore, it can be further assumed that the flow at each longitudinal section of the waverider is independent.
- 3) The angles between the freestream flow and different longitudinal profiles of the lower surface are slightly different, making it difficult to deal with the effect of angle of attack. Thus, based on the above

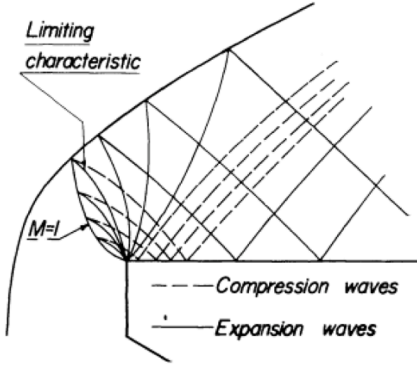


Fig. 14. Illustration of interaction of different waves [43].

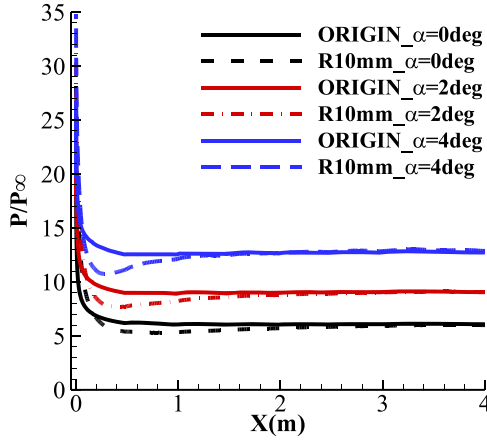


Fig. 15. Pressure distribution along the centerline of the lower surface.

two assumptions, the waverider is finally simplified as the flat plate with variable sweep angle shown in Fig. 22.

3.2. Model construction

Based on the above flat plate, the model can be constructed from the computational data obtained at different longitudinal sections. Note that the pressure increment is calculated by subtracting the pressure on the sharp flat plate from that on the blunted flat plate ($R = 10$ mm) at the corresponding position.

1) Determine the key physical parameters

The flight parameters M_∞ , H , and α can all affect the induced pressure on the upper and lower surface of the blunted flat plate. Fig. 23 compares the distribution of the non-dimensional pressure increment along the centerline of the lower surface at different flight altitudes. It's found that only minor difference exists and the difference decreases

gradually as the angle of attack increases. In fact, as shown in Fig. 21, the streamline of the waverider itself has a certain deflection angle (about 6.3° for CW3). Hence, the influence of flight altitude, namely the viscous interaction effect, on the induced pressure increment can be ignored.

In addition, the pressure increment distribution of the upper surface is plotted in Fig. 24. Unlike the phenomenon of the lower surface, there is no significant difference between the pressure increments at different angles of attack for the upper surface. Thus, it can be approximated that the pressure increment distribution at different angles of attack is equal to that at $\alpha=0^\circ$.

Based on the above analysis, the flight conditions listed in Table 4 are chosen here to construct the pressure increment database for SR.

Furthermore, based on the previous study [42], the geometric parameters affecting the induced pressure are Λ and x/R for each longitudinal cross-section of the flat plate. Thus, the non-dimensional pressure increment can be expressed by the following function form:

$$\Delta P / P_\infty = f(M_\infty, \alpha, \Lambda, x/R) \quad (3)$$

The flat plates FP1 and FP3, shown in Fig. 22, are taken here to train the target model. From Fig. 25 we can find that the two flat plates cover a large range of sweep angle and have good representativeness. Then the pressure increment on all the longitudinal cross-sections, which correspond to z -coordinate of the symbols in Fig. 25, is chosen to build the training database.

1) Determine the mathematical operators

In the process of SR, the effective selection of mathematical operators can not only reduce the complexity of equations, but also be easier to capture the underlying physical mechanism. Thus, the mathematical operators are selected here by referring to the existing theories that can capture the partial influence of the bluntness effect on pressure distribution. These theories are as follows:

1) Blunt-nosed flat plate (first approximation) [27]:

$$P / P_\infty = 0.121 M_\infty^2 C_D^{2/3} (x/d)^{-2/3} \quad (4)$$

2) Spherically blunted cylinder (first approximation) [27]:

$$P / P_\infty = 0.067 M_\infty^2 C_D^{1/2} (x/d)^{-1} \quad (5)$$

3) Spherically blunted cone [28]:

$$\begin{cases} P = P_C - (P_C - P_2)e^{-\eta} \\ \eta = \left(\frac{\delta P}{\delta S} \right)_2 \frac{(x - x_2)}{(P_C - P_2)\cos\delta_2} \end{cases} \quad (6)$$

Note that Eqs. (4)–(5) can predict the blast-wave effect and Eq. (6)

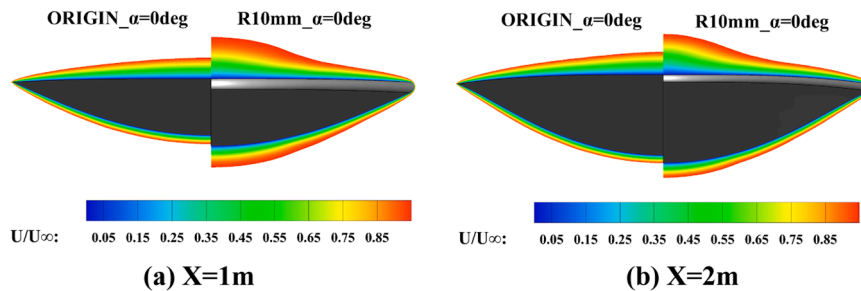


Fig. 16. Comparison of velocity contour between the original and blunted waveriders.

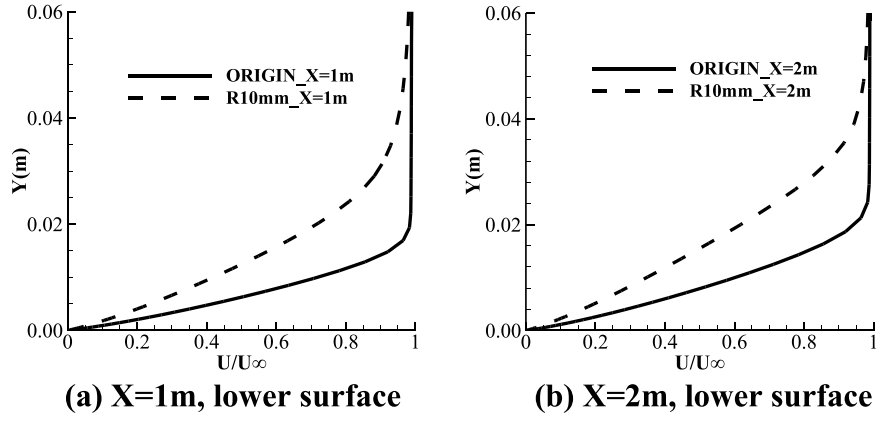


Fig. 17. Comparison of velocity profile between the original and blunted waverider.

can predict the more complex pressure overexpansion effect near the shoulder of a spherically blunted cone.

Therefore, based on the above theories, the operator set is determined initially as $\{+, -, \times, /, \sin, \cos, \exp\}$.

3) GA optimization

According to the independent variables, mathematical operators, and the pressure increment database obtained above, SR can be conducted along with the GA optimization. The detailed framework is given in Fig. 26. The initial candidate symbolic functions are generated by the random combination of independent variables and mathematical operators. Based on the errors of these equations, best functions are selected.

Then, new candidate functions are generated by the small variation of best functions through selection, crossover, and mutation. The above process is repeated until sufficient accuracy or the last generation is reached.

Based on the above procedure, the final model of pressure increment obtained by SR is as follows:

$$f = \frac{(0.18 \cdot \cos \Lambda + 0.37 \cdot \sin \alpha) M_\infty^2}{2.14 + x/R} + \frac{-0.85 \cdot \cos \Lambda \cdot \sin^2 \alpha \cdot M_\infty^2}{\exp(0.41 \cdot \sin^2 \alpha \cdot x/R)} + \frac{-0.046 \cdot M_\infty^2}{\exp(1.34 \cdot \cos \Lambda \cdot x/R)} \quad (7)$$

where M_∞ is freestream Mach number, α is angle of attack, Λ is local

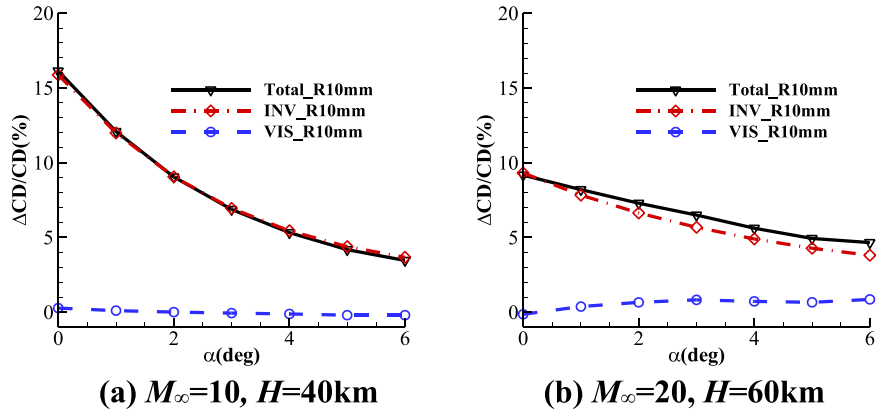


Fig. 18. Comparison of the drag variation for different flight conditions, $R = 10$ mm.

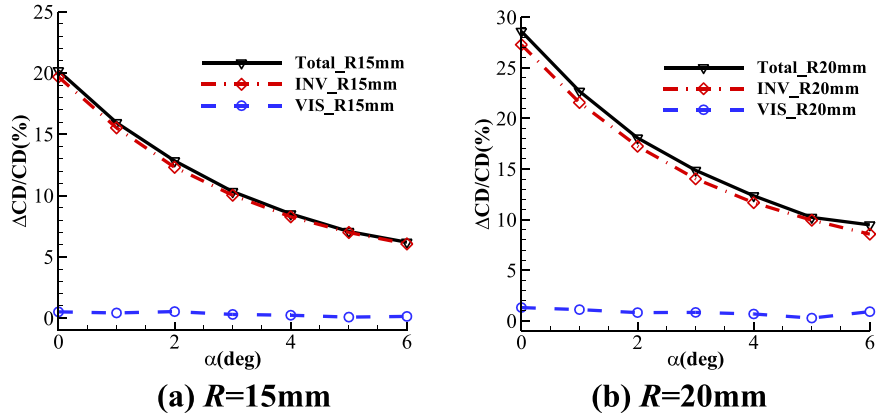


Fig. 19. Comparison of the drag variation for different bluntness radii ($M_\infty=15$, $H = 50$ km).

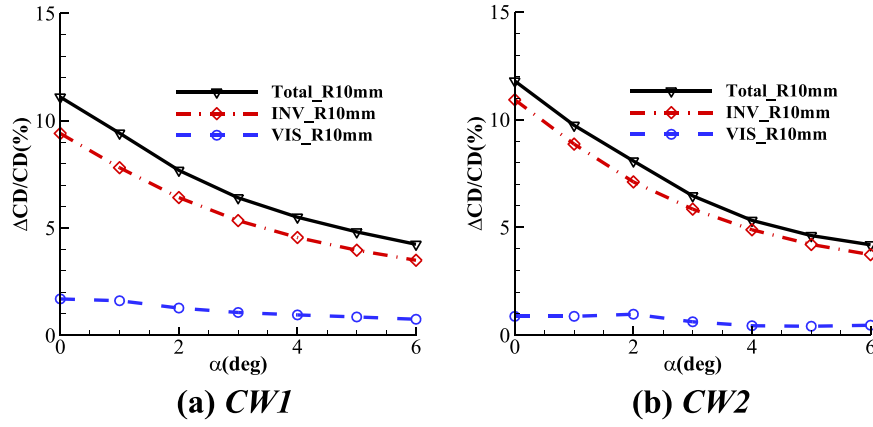


Fig. 20. Comparison of the drag variation for different waveriders ($M_\infty=15$, $H=50$ km).

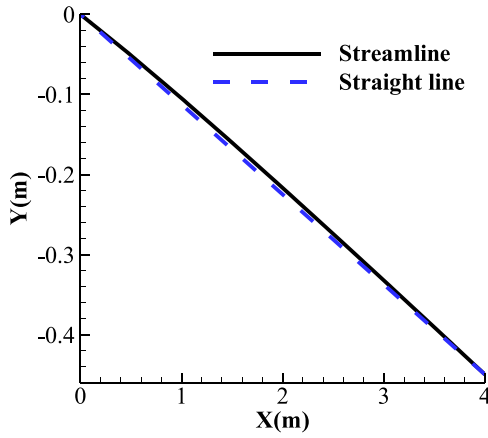


Fig. 21. Comparison of different lines.

sweep angle of the leading edge, x and R is distance measured from the nose in the flow direction and bluntness radius of the leading edge, respectively.

It can be found from the function that the first term is first-order monotonically decreasing equation for $x/R > 0$. The second and third term increases exponentially as x/R increases. Compared with the second term, the magnitude of the third term decreases significantly as x/R increases because it has a larger exponential coefficient.

In order to explain the physical implication of the model, taking the longitudinal section ($\Lambda=40^\circ$) of $FP3$ at $M_\infty=15$, $\alpha=6^\circ$ as an example, Fig. 27 compares the pressure increment distribution obtained from CFD and the analytical model, where the prediction of each single term in Eq. (7) is also decomposed and plotted. It can be found that the first term, which is similar to the form of Eq. (5) for $\Lambda=0^\circ$ and $\alpha=0^\circ$, can mainly capture the blast-wave effect. And according to the pressure distribution, the second term mainly corresponds to the pressure overexpansion effect. In addition, the third term can mainly capture the local pressure peak near the very nose caused by the sweep angle, which can be further observed in Fig. 28. Therefore, the combined effects of the three terms make the prediction of the model agree well with the original training data.

The flat plate $FP2$ (shown in Fig. 22) and original waverider $CW2$ are used to validate the accuracy of the model. The results are compared in Figs. 28 and 29 at different flight conditions, showing that the predicted data is in good agreement with CFD results for both the flat plate and original waverider. Note that for some cross sections with large sweep angles ($\alpha \geq 0.4$), the result is somewhat underestimated by the approximate model. However, such difference is only restricted to the region very close to the nose and affects little to the total forces.

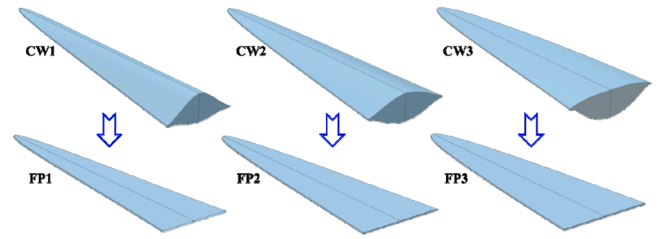


Fig. 22. Illustration of geometry simplification.

Therefore, based on the above analysis, the analytical model constructed by SR can be used to capture the pressure increment of the waverider induced by the detached shock wave with reasonable accuracy.

4. Waverider optimization framework

4.1. Aerodynamic model

Aerodynamic models are widely used in various kinds of hypersonic aerodynamic optimization during the preliminary design due to the high efficiency. For obtaining the aerodynamic forces of blunted waveriders, this paper first calculates the distribution of the pressure and shear stress on the original sharp waveriders. Then, based on the assumption that the variation of friction drag caused by bluntness can be neglected, only the pressure on the blunted leading edge and the pressure increment on the upper and lower surface need to be evaluated. Finally, the CL , CD , L/D , and Xcp are calculated by the integration of local surface pressure and shear stress on the blunted waverider. The detailed aerodynamic models used in this paper are as follows.

An improved Tangent-cone method [45] is employed to approximate the inviscid pressure on the windward surface and the hypersonic expansion-wave relation is used on the leeward surface [42]:

$$p = \begin{cases} 1 + \left(\frac{2\gamma}{\gamma+1} M_\infty^2 \sin^2 \beta - \frac{\gamma-1}{\gamma+1} \right) \cdot \left(1 + \frac{\gamma M_\infty (\beta - \theta)^2 \cos^2 \beta}{1 + [(\gamma-1)/2] M_\infty^2 \sin^2 \beta} \right)^{-1} & (K \geq 0) \\ \left(1 + \frac{\gamma-1}{2} K \right)^{\frac{2\gamma}{\gamma-1}} & \left(-\frac{2}{\gamma-1} < K < 0 \right) \\ 0 & \left(K \leq -\frac{2}{\gamma-1} \right) \end{cases} \quad (8)$$

where $p=p_w/p_\infty$, the hypersonic similarity parameter $K = M_\infty \theta$. The shock wave angle β can be approximated as [46]:

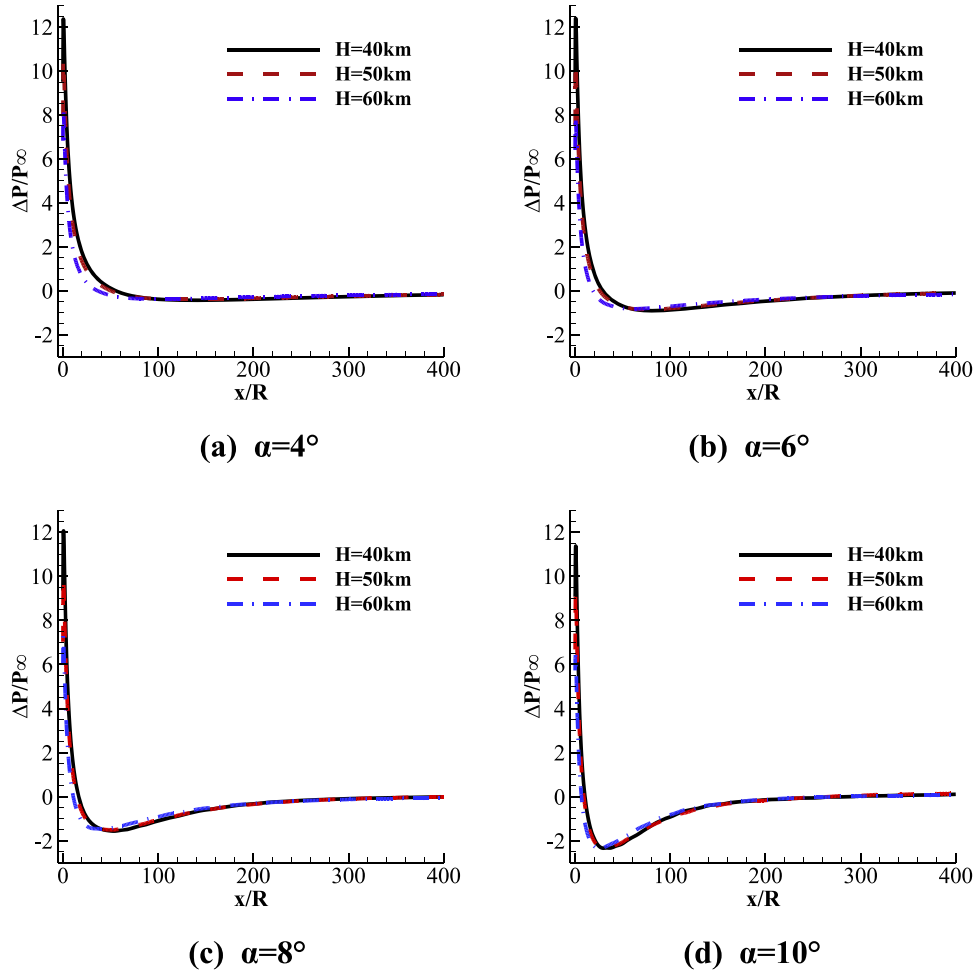


Fig. 23. Pressure increment distribution along the centerline of lower surface, $M_\infty=15$.

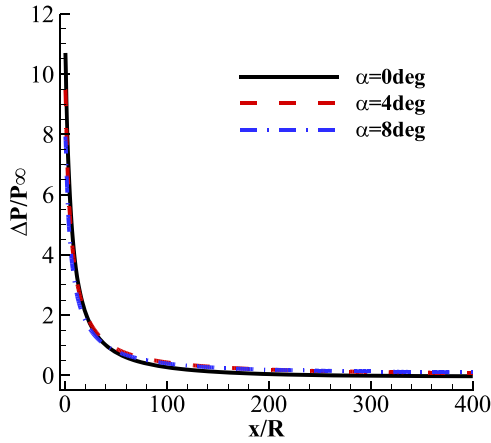


Fig. 24. Pressure increment distribution along the centerline of upper surface, $M_\infty=15$, $H = 50$ km.

$$\beta = \frac{\gamma + 1}{\gamma + 3} \left[1 + \sqrt{1 + \frac{2(\gamma + 3)}{(\gamma + 1)^2 M_\infty^2 \theta^2}} \right] \cdot \theta \quad (9)$$

The engineering approach put forward by Bertram [47] is adopted to obtain the boundary layer displacement thickness rapidly, which was also employed by Anderson [26] and Liu [14] in the waverider optimization. The expression is as follows:

Table 4

Flight conditions.

H	M_∞	ΔM_∞	α	$\Delta \alpha$
50km	[10,20]	2	$[0^\circ, 16^\circ]$	2°

$$\frac{d\delta_x^*}{dx} = \frac{\xi}{M_\infty \sqrt{p}} \left(1 + \frac{\xi}{2p} \frac{dp}{d\xi} \right) \quad (10)$$

where,

$$\begin{cases} \xi = 0.425(\gamma - 1) \left(\frac{T_w}{T_{ad}} + 0.35 \right) \bar{\chi} \\ \bar{\chi} = \frac{M_\infty^3}{\sqrt{\text{Re}_{x,\infty}}} \sqrt{C_w} \\ C_w = \left(\frac{T^*}{T_\infty} \right)^{\frac{1}{2}} \frac{T_\infty + 110.4}{T^* + 110.4} \end{cases} \quad (11)$$

where the subscript ∞ and w represents the quantities based on undisturbed freestream conditions and wall conditions, respectively; T_{ad} is the adiabatic wall temperature and T^* is the reference temperature, which can be approximated by the following expressions [48]:

$$\frac{T^*}{T_\infty} = 1.28 + 0.023 M_\infty^2 + 0.58 \left(\frac{T_w}{T_\infty} - 1.0 \right) \quad (12)$$

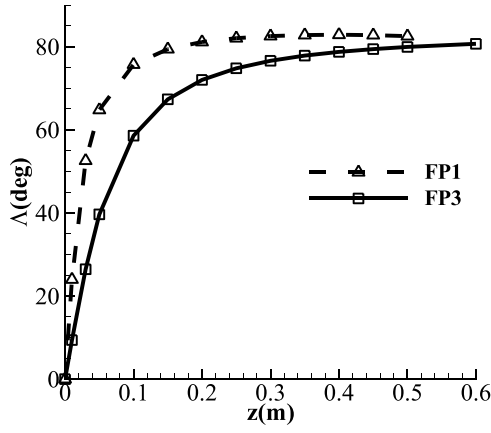


Fig. 25. The distribution of sweep angle along the leading edge.

$$\frac{T_{ad}}{T_{\infty}} = 1 + \sqrt{\text{Pr}^*} \cdot \frac{\gamma - 1}{2} M_{\infty}^2 \quad (13)$$

where $\text{Pr}^* = 0.72$, and $T_w = 1000$ K are assumed.

The modified Newtonian theory [42] is applied to calculate the pressure on the blunted leading edge:

$$C_p = C_{p\max} \sin^2 \theta \quad (14)$$

where,

$$C_{P\max} = \frac{2}{\gamma M_{\infty}^2} \left(\frac{P_{O_2}}{P_{\infty}} - 1 \right)$$

$$\frac{P_{O_2}}{P_{\infty}} = \left[\frac{(\gamma + 1)^2 M_{\infty}^2}{4\gamma M_{\infty}^2 - 2(\gamma - 1)} \right]^{\gamma/(\gamma-1)} \left[\frac{1 - \gamma + 2\gamma M_{\infty}^2}{\gamma + 1} \right] \quad (15)$$

Finally, the model Eq. (7) is employed to evaluate the pressure increment on upper and lower surfaces induced by the detached shock wave.

The detailed steps for calculating the aerodynamic forces are summarized as follows:

- 1) Calculate the pressure distribution on the original body surface using Eq. (8);
- 2) Calculate the viscous interaction parameters $\bar{\chi}$ and ξ by Eqs. (11)–(13);
- 3) Combine Eqs. (8) and (10) to obtain an ordinary differential equation about $dp/d\xi$, which is then solved by the fourth-order Runge-Kutta method starting at the leading edge and progressing in the X-direction with the initial conditions: $\bar{\chi} = 0$, $p = p_{w0}/p_{\infty}$, where p_{w0} is the pressure calculated in step (1).
- 4) The laminar skin friction is evaluated by the following formula which can consider the strong viscous interactions [49]:

$$C_f = 0.664 \sqrt{\frac{p(x)C_w}{\text{Re}_{x,\infty}}} \quad (16)$$

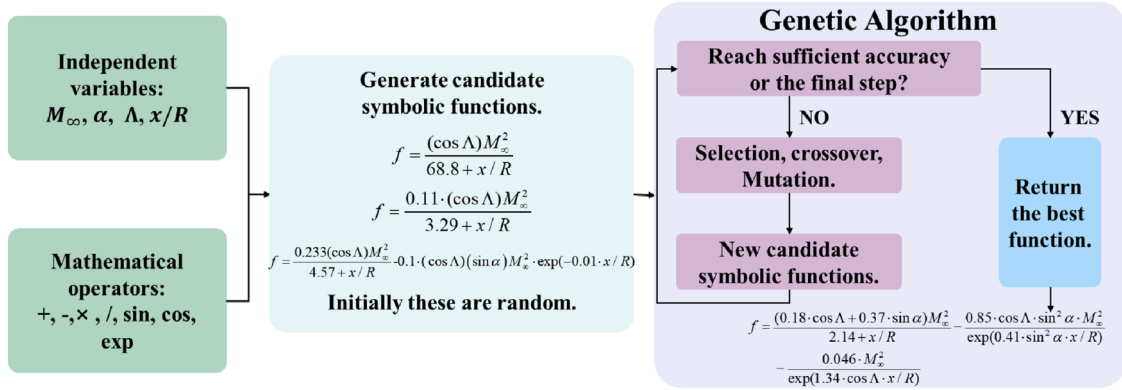


Fig. 26. SR framework.

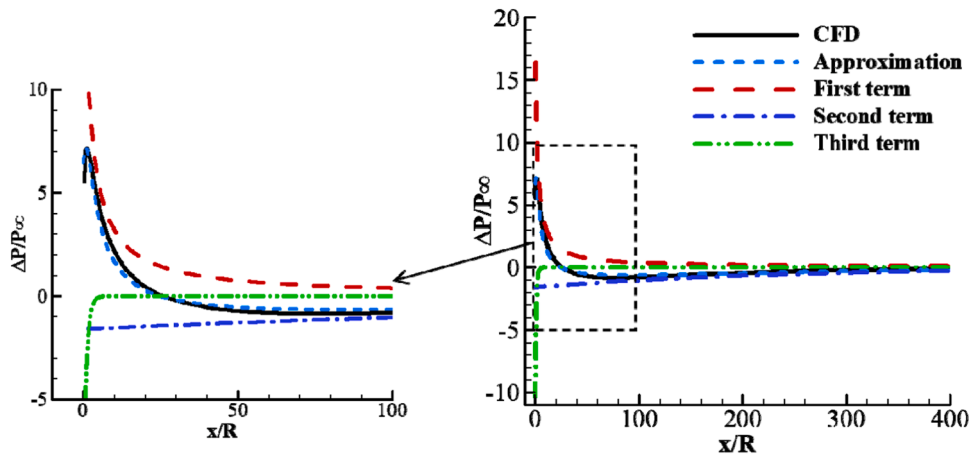


Fig. 27. Comparison of pressure increment distribution.

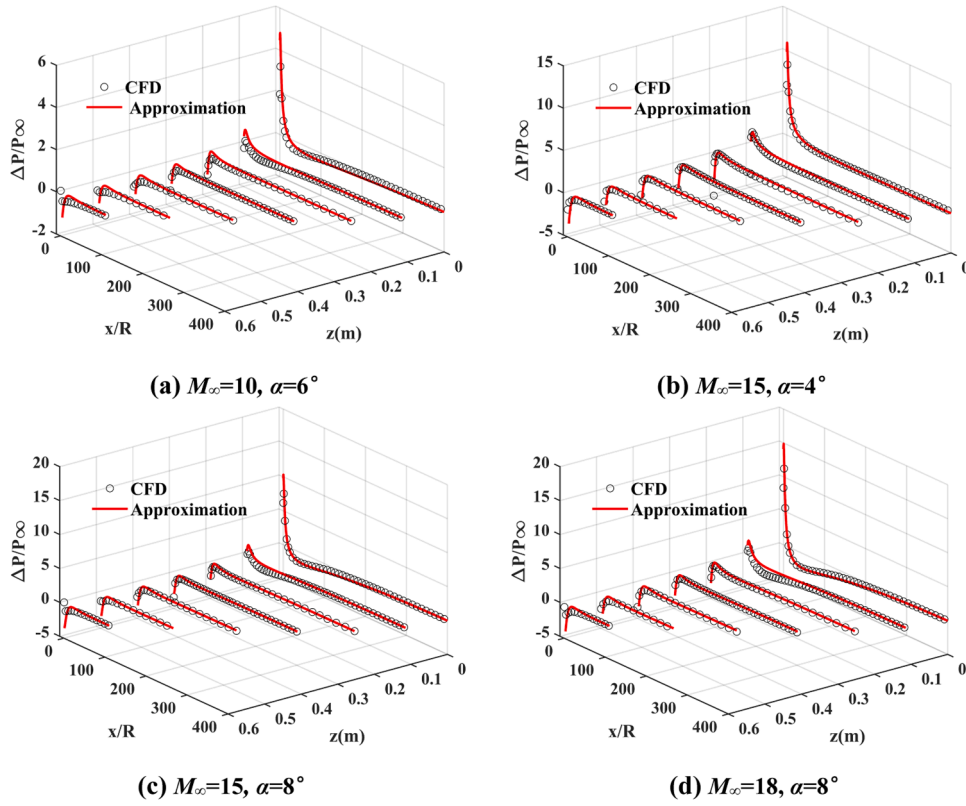


Fig. 28. Comparison of the pressure increment distribution on FP2.

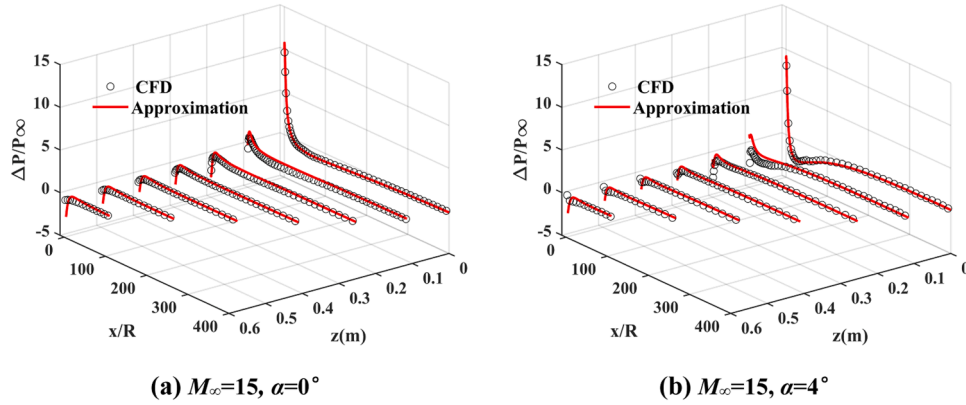


Fig. 29. Comparison of the pressure increment distribution on CW2.

- 5) Calculate the pressure on the blunted leading edge via Eqs. (14)–(15).
- 6) Calculate the pressure increment by Eq. (7), and then obtain the final pressure on the upper and lower surfaces.
- 7) The base pressure is approximated by the freestream static pressure.
- 8) Obtain the final aerodynamic performance by the integration of forces on body surface.

Taking the blunted waverider CW2 ($R = 10$ mm) as an example, Fig. 30 compares the aerodynamic forces calculated by CFD and the aerodynamic model (AeroModel). It can be found that the results from the aerodynamic model match the CFD data quite well, with the maximum relative difference of CL , CD , L/D , and X_{cp} being only 5.46 %, 3.61 %, 2.53 %, and 0.76 %, respectively. This is acceptable for the rapid aerodynamic optimization during the preliminary design stage.

4.2. Genetic algorithm optimization

The GA is also adopted here and the optimization framework of the blunted waverider is given in Fig. 31, where the trim constraint is included. If no constraint is assigned, the program will directly pass the corresponding step to the next one. The leading edge design parameters kw , φ , and η are set to be the optimization variables. Note that for the optimization of the original waverider with sharp leading edge, the pressure relevant to the bluntness is not taken into account. After the design Mach number, flight altitude, shock wave angle, bluntness radius, and the constraint are specified, the optimum waverider can be obtained through the optimization framework.

5. Results and analysis

The current study is focused on the design condition of $M_\infty=15$ and

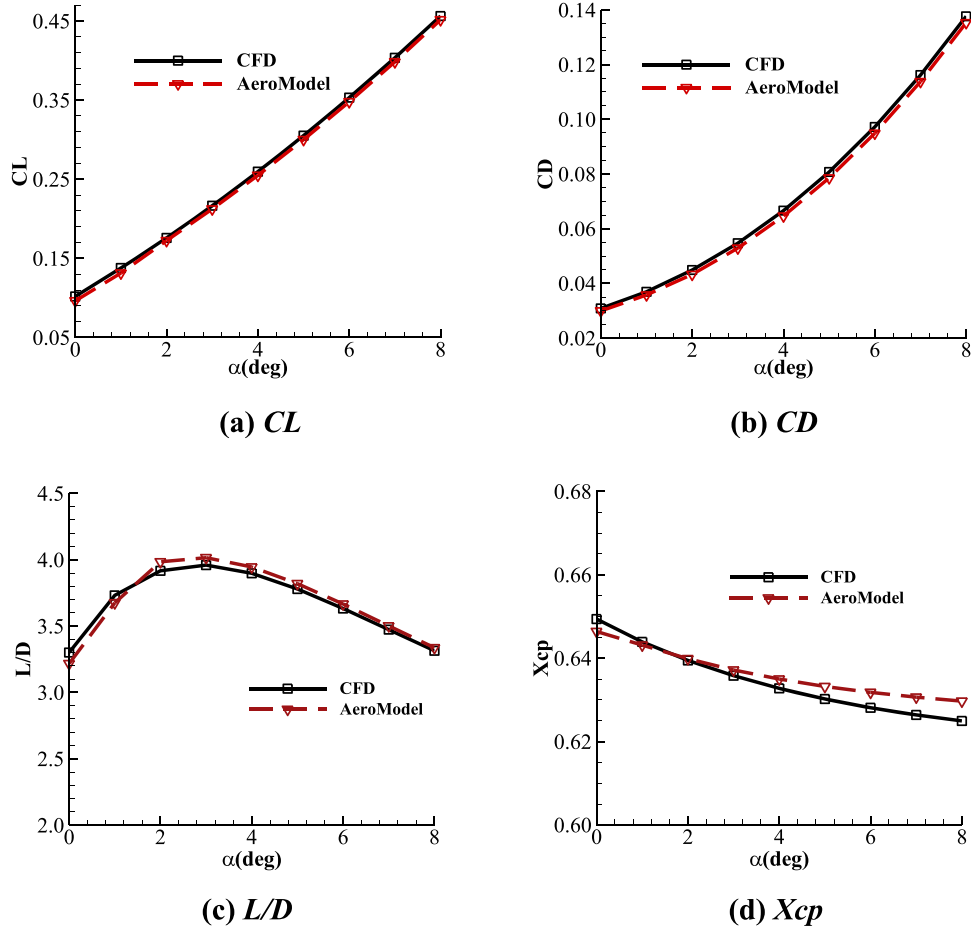


Fig. 30. Comparison of aerodynamic forces between CFD and aerodynamic model.

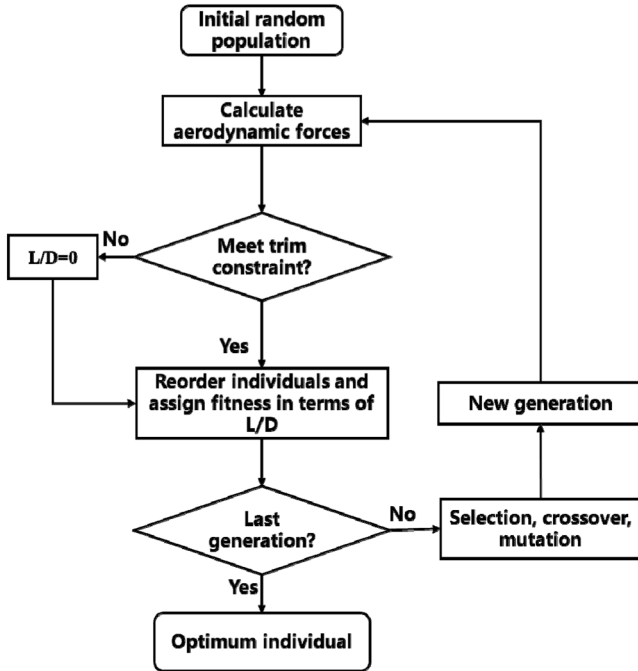


Fig. 31. GA optimization framework of the blunted waverider.

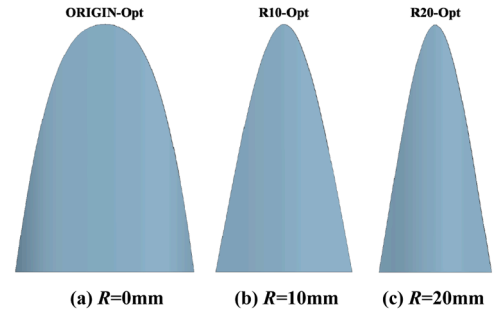


Fig. 32. Optimum waveriders with different bluntness radii.

$H = 50$ km. The optimum waveriders with both the sharp and blunted leading edge are generated. Then all the configurations are blunted with corresponding leading edge radius, and the aerodynamic performance calculated by CFD is compared. In practical applications, enough lift is required to balance the weight of the vehicles. Therefore, the L/D for given lift coefficient is taken as the optimization objective.

First, the optimum waveriders with different bluntness radii for $CL=0.4$ are obtained. These configurations are shown in Fig. 32. Significant differences exist among the three waveriders: as the bluntness radius increases, the optimum waverider turns narrower and the sweep angle at the same cross section around the nose turns larger, as shown in Fig. 33.

The L/D of different optimum waveriders for the same bluntness radius is shown in Fig. 34. Compared to the blunted waverider *ORIGIN*-

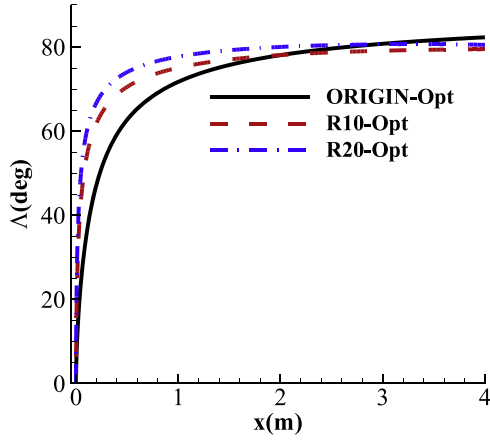


Fig. 33. Comparison of sweep angle for the three waveriders.

Opt, higher aerodynamic efficiency of the blunted optimized waverider *R10-Opt* and *R20-Opt* is obtained, with the design L/D corresponding to $CL=0.4$ being improved by 12.0 % (*R10*: from 3.50 to 3.92) and 27.40 % (*R20*: from 2.81 to 3.58), respectively. The main reason can be explained as follows. It's discussed in Section II that the increase of total drag caused by bluntness is mainly due to the wave drag of the blunted leading edge. And as shown in Fig. 35, the larger sweep angle near the

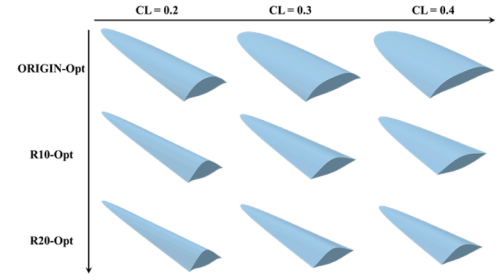
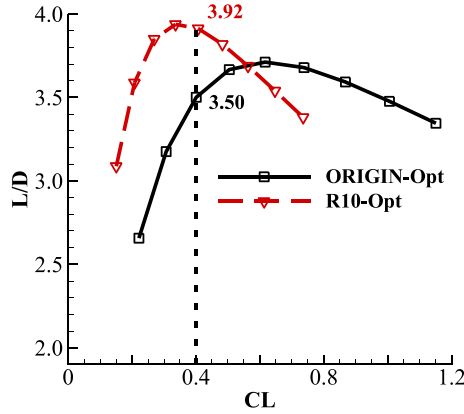


Fig. 36. Optimum waveriders for different lift coefficients and bluntness radii.

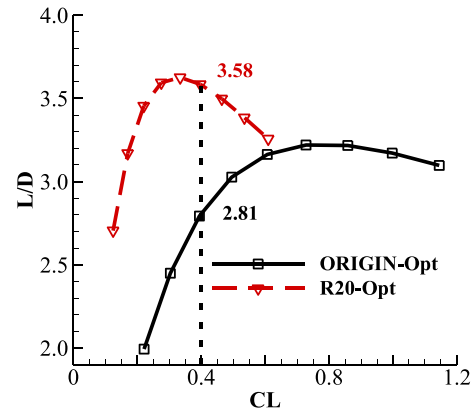
nose of the blunted optimized waverider produces lower wave drag of the blunted leading edge.

Furthermore, by specifying different lift coefficient requirement, a series of optimum waveriders with different bluntness radii are obtained through optimization, as shown in Fig. 36. Not surprisingly, the optimum configuration turns wider for higher lift coefficient requirement. The design L/D of the blunted waveriders is compared in Fig. 37 and the relative improvement of the blunted optimized waveriders is also plotted in Fig. 38. It shows that all the blunted optimized waveriders, *R10-Opt* and *R20-Opt*, have apparently higher design L/D than *ORIGIN-Opt* for different lift coefficient constraint. And the improvement of L/D becomes larger as the lift coefficient or bluntness radius increases.

In addition, by changing the design Mach number, a series of

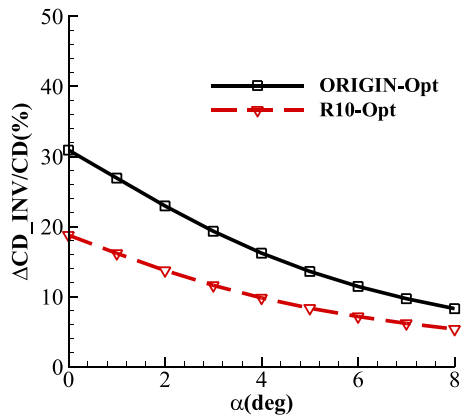


(a) $R=10\text{mm}$

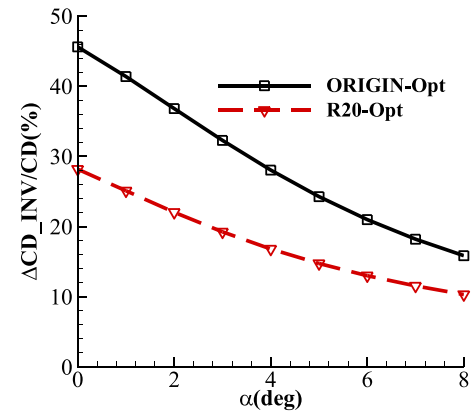


(b) $R=20\text{mm}$

Fig. 34. Comparison of L/D between different blunted waveriders.



(a) $R=10\text{mm}$



(b) $R=20\text{mm}$

Fig. 35. Comparison of wave drag of the blunted leading edge.

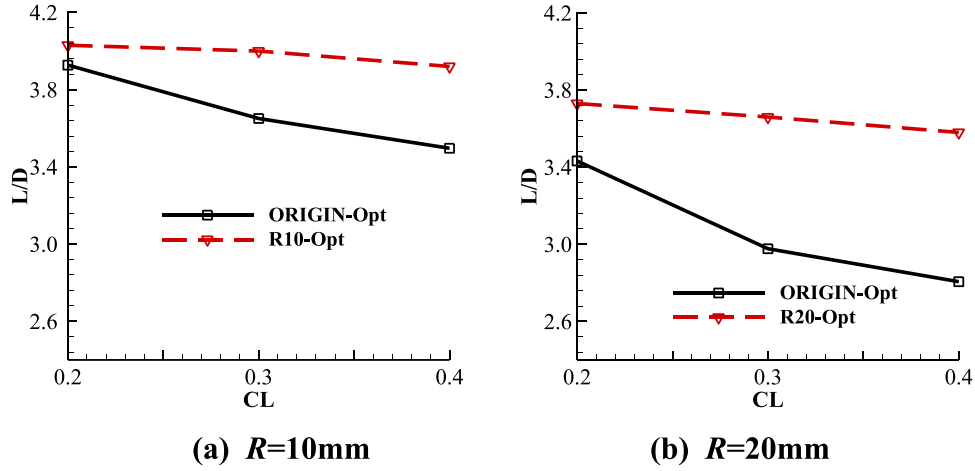
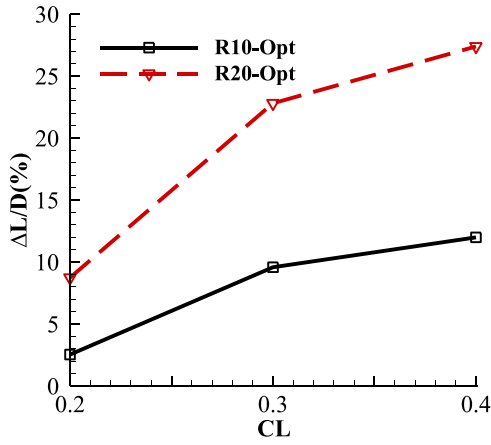
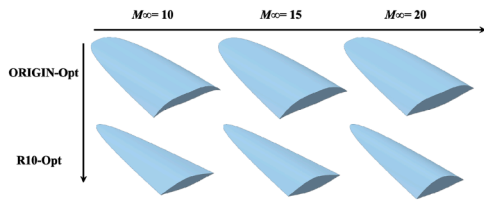
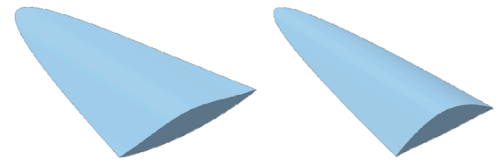
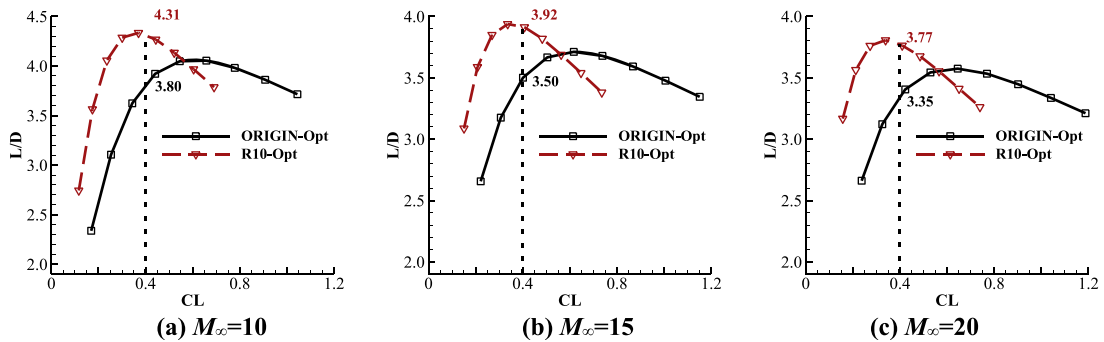
Fig. 37. Comparison of design L/D of the optimum waveriders.Fig. 38. Relative improvement of design L/D for the blunted optimized waveriders.

Fig. 39. Optimum waveriders for different design Mach number.

Fig. 41. Comparison of optimum waveriders with the constraint $X_{cp}=0.62$.Fig. 40. Comparison of L/D between different blunted waveriders ($R = 10\text{mm}$).

optimum waverider at the different flight condition are obtained, as demonstrated in Fig. 39. The L/D of the optimum waverider after bluntness is compared in Fig. 40. It can be found that compared to the blunted waverider ORIGIN-Opt, the improvement of the design L/D of R10-Opt is more than 12.0 % at the different design Mach numbers. Therefore, within the range of Mach number studied in this paper, the optimization algorithm can significantly improve the design L/D of the blunted optimum waverider.

In addition to L/D , trim is another key factor that must be considered during the preliminary design of any practical vehicle. In order that no extra trim drag is produced by the elevator at the design condition, the constraint of trim is added to the optimization process, requiring that the center of pressure be coincident with the center of gravity.

Taking the constraint $X_{cp}=0.62$ (also $X_{cg}=0.62$) for $CL=0.4$ as an example, the optimum waveriders with sharp and blunted leading edge are shown in Fig. 41 and the aerodynamic performance is plotted in Fig. 42, including L/D , X_{cp} , and Cm_z . Note that referring to standard convention, a positive pitching moment corresponding to a nose-up moment is ruled here. First, the design L/D is improved by 4.80 % for the blunted optimized waverider. Moreover, the X_{cp} is very close to the

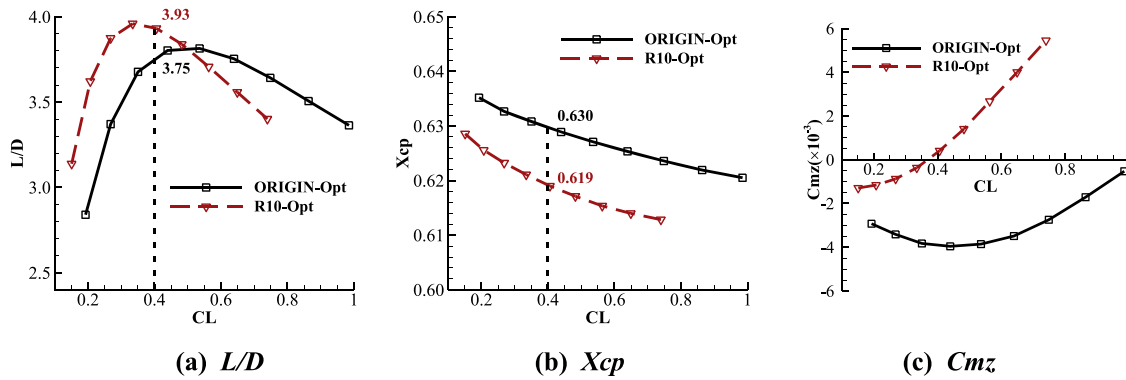


Fig. 42. Comparison of aerodynamic performance between different blunted waveriders ($R = 10$ mm).

center of gravity, meaning that the configuration is almost self-trimmed as shown in Fig. 36(c). However, the trim characteristic is far away from requirement for the blunted waverider *ORIGIN-Opt*. Accordingly, the advantage of blunted optimized waverider is also significant in terms of the overall performance, including both the L/D and the trim characteristic.

6. Conclusions

The influence of leading edge bluntness on the aerodynamic performance of the original waverider is investigated in this paper. It's found that the total drag increases and the total lift decreases significantly, which is mainly attributed to the wave drag of the blunted leading edge and the lift variation of the upper and lower surface affected by the detached shock, respectively. It's also found that the friction drag of the blunted leading edge can almost be counteracted by the friction drag reduction of the upper and lower surface, in that the slope of the velocity profile inner the boundary layer turns obviously lower after bluntness. The result is that the L/D is much reduced. In addition, the X_{cp} also moves backward evidently after bluntness, which has vital influence on the trim characteristic. Further analysis suggests that the pressure overexpansion effect is the main reason for the variation of the lift and X_{cp} .

Furthermore, the methodology of symbolic regression is adopted to construct an analytical pressure increment model of the upper and lower surface caused by the bluntness effects. In combination with other traditional approximate models, including the Tangent-cone method, modified Newtonian theory, boundary layer displacement thickness determination method, and laminar skin friction formula, an efficient and accurate aerodynamic model is constructed, which can consider the impacts of leading edge bluntness. Then a Genetic Algorithm optimization framework for the blunted waveriders is presented based on the aerodynamic model.

The optimum waveriders with both the sharp and blunted leading edge are generated and compared with the corresponding leading edge bluntness. For the given lift coefficient requirement, evident differences exist for different optimum waveriders: as the bluntness radius increases, the optimum waverider turns narrower and the sweep angle at the same cross section around the nose turns larger. Therefore, the wave drag of the blunted optimized waverider is reduced greatly and the L/D is much higher than that of the blunted waverider optimized without considering the bluntness effects. It's also found that the improvement of L/D turns larger as the bluntness radius or the design lift coefficient increases. Finally, when the X_{cp} is constrained during the optimization, the resulting blunted optimized waverider exhibits both better trim characteristic and higher L/D .

In summary, compared with the traditional waverider optimization, the optimization process for the blunted waveriders results in distinctly different shapes with better aerodynamic performance. Therefore, the

direct incorporation of bluntness effects into the optimization process is quite necessary. In addition, the optimization framework of the blunted waveriders may also be expanded to the design of other kinds of hypersonic configurations.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

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